

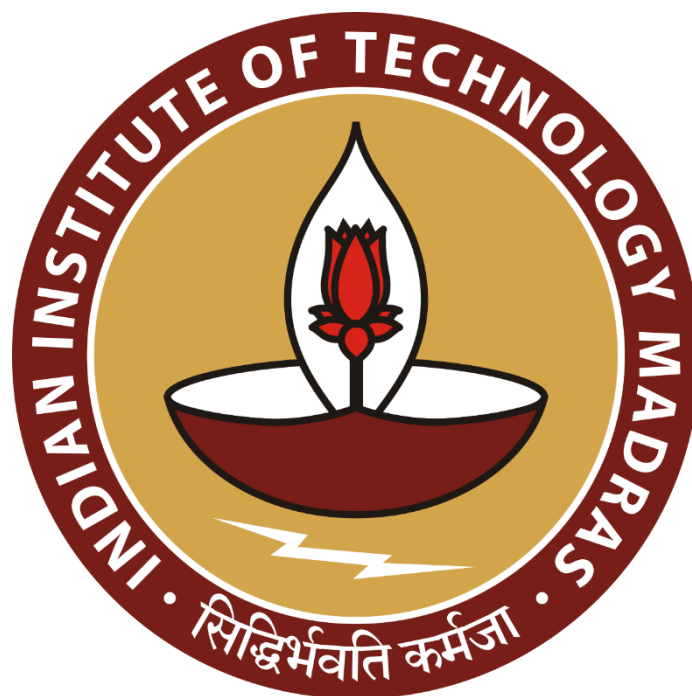
Data-Driven Optimization and Scalability of Diabetes Care Medical and General Stores in Retail Pharmacy sector

A Proposal report for the BDM capstone Project

Submitted by

Name: Arya Mulay

Roll number: 23F3002123



IITM Online BS Degree Program,
Indian Institute of Technology, Madras, Chennai
Tamil Nadu, India, 600036

Contents

1	Executive Summary and Title	3
2	Organisation Background	3
3	Problem Statement (Listed as objectives)	4
3.1	Problem statement 1	4
3.2	Problem statement 2	4
4	Background of the Problem	4
5	Problem Solving Approach	4-5
6	Expected Timeline	6-7
7	Expected Outcome	7

Declaration Statement

I am working on a Project titled “**Data-Driven Optimization and Scalability of Diabetes Care Medical and General Stores in Retail Pharmacy sector**”. I extend my appreciation to **Diabetes Care Medical and General Stores**, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfillment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.

Signature of Candidate:



Name: Arya Mulay

Date: 13-05-2025 (13th May 2025)

1 Executive Summary and Title

Project Title: Data-Driven Optimization and Scalability of Diabetes Care Medical and General Stores in Retail Pharmacy sector

Organization Background: Diabetes Care Medical and General Stores is a trusted retail pharmacy that began in 2010 to support the patients of the Diabetes Care Center started in 2002 in Solapur, Maharashtra. The main aim was to provide quality and affordable diabetic care products like insulin vials and syringes, as well as general medicines, while educating patients on proper usage and care.

About the Problem: Even though the store earns good profits and has loyal customers, it's finding it hard to open a new branch due to lack of proper financial planning and tracking. Also, the pharmacy still uses outdated software for inventory, which causes problems like medicines getting overstocked or running out at the wrong time.

Possible Approach and Solution: The project focuses on two key areas:

First, using profit and supplier credit data to model whether the pharmacy can safely open another branch. Second, improving inventory management through sales data analysis, ABC analysis, and demand forecasting to make smarter stock decisions and reduce waste and stockouts.

Expected Outcome: These methods will help the pharmacy plan better financially, avoid stock issues, and maybe grow their business. It will also make everyday work easier for staff and improve customer satisfaction.

2 Organization Background

Diabetes Care Medical and General Stores has been well established since 2010 with an objective to address the challenges faced by diabetic patients from the main center clinic Diabetes Care Center first established in 2002. At that time, patients were often provided costly substitutes for prescribed drugs and poor-quality syringes and insulin vials. Recognising the problems faced by the patients, the center clinic in 2010 formed a separate pharmacy retail firm Diabetes care Medical and General Stores with a motto of giving quality affordable medicine with proper education.

Today, Diabetes Care Medical and General Stores specializes in diabetes care products and general medicines, ensuring that patients receive genuine, affordable medications and essential supplies like insulin vials, syringes, and glucose monitoring devices. The organization prioritizes maintaining the integrity of the cold chain for temperature-sensitive products and provides prescribed patients on proper medication usage and diabetes management along with general healthcare.

As of now, the organization has become a trusted name in the pharmacy sector for the well-being of its patients.

3 Problem Statement (Listed as objectives)

3.1 Problem statement 1:

The Diabetes Care Pharmacy is finding it difficult to scale up its business by opening additional retail locations, despite having sufficient profits and strong supplier relationships, due to limitations in capital, operational capacity and not using the supplier credit for expansion.

3.2 Problem statement 2:

The Diabetes Care Pharmacy struggles to maintain optimal inventory levels, leading to frequent overstocking or stockouts, due to limited inventory control and reliance on outdated record-keeping software.

4 Background of the Problem

While discussing with the two partners of the retail pharmacy firm, I discovered the two major problems mentioned above:

The Diabetes Care Pharmacy has been doing well since it started in 2010, with monthly profit margins between 30% and 40%. It has good sales and a loyal set of customers, which makes it a strong candidate for expanding the business. But even with all this, the pharmacy finds it hard to open a new branch. The main issue is getting the money needed to start, since there's no proper system to track finances or cash flow automatically. Because of this, it's tough to plan how to use profits or take advantage of the credit given by suppliers to grow the business safely.

The Diabetes Care Pharmacy still uses old software called Green Care and mostly relies on manual work to track sales and inventory. The system doesn't show real-time stock levels, doesn't give alerts, and isn't connected to the sales data properly. Because of this, staff usually have to check stock by hand and make corrections after sales, which often leads to mistakes. When we looked at the sales data from the last 10 months, we saw a lot of issues. Some medicines were overstocked and got wasted, while others ran out, causing missed sales and unhappy customers. Not having a proper data-based system has made it hard for the pharmacy to manage inventory well and keep things running smoothly.

5 Problem Solving Approach

Methods:

For Problem Statement 1, I plan to use basic financial modeling to understand if the current profit margins and ask for supplier credit can support opening a second store. I'll go through

monthly profit numbers and check how much money can be reused for investment. Since suppliers may give 20–25 days of credit based on the huge profit margins of the pharmacy store. I would run simulations in Excel to see if the business can handle the cash flow and still make enough profit to safely expand.

For Problem Statement 2, I would work on inventory optimization. First, I'll use ABC analysis to group products based on how often they sell and how important they are. Then I'll apply demand forecasting using past sales data(10 months) to predict which products need to be restocked when. I'll also calculate things like safety stock and reorder levels so that the pharmacy can avoid both overstocking and running out of important medicines. The goal is to create a system that helps them make smarter inventory decisions using data instead of guesswork.

Data:

For Problem 1, I'll be using financial data from April 2024 to January 2025. This includes monthly sales, profit margins (which are around 30–40%), supplier credit terms, and expense details. I'll focus mainly on the cash flow and how much of the profit is available for reinvestment, along with credit period usage to reduce initial costs.

For Problem 2, I'll be working with sales data of 50 commonly sold items like insulin vials, syringes, and general medicines. I'll analyze stock movement over 10 months, including purchase records and manual adjustments. I'll use this to find which items are overstocked or often out of stock and try to improve that using proper demand-based planning.

Tools:

I'll use Excel or Google Sheets for most of the modeling and analysis. They're easy to use and good for quick insights. For long-term fixes, especially for problem 2, I would suggest using an ERP tool like Tally, Marg ERP, or Zoho Inventory to automate their finances and inventory, instead of old and outdated database software like Green Care.

6 Expected Timeline

6.1 **Work Breakdown Structure:** The Work Breakdown Structure is given as follows in the below mentioned Figure 1.

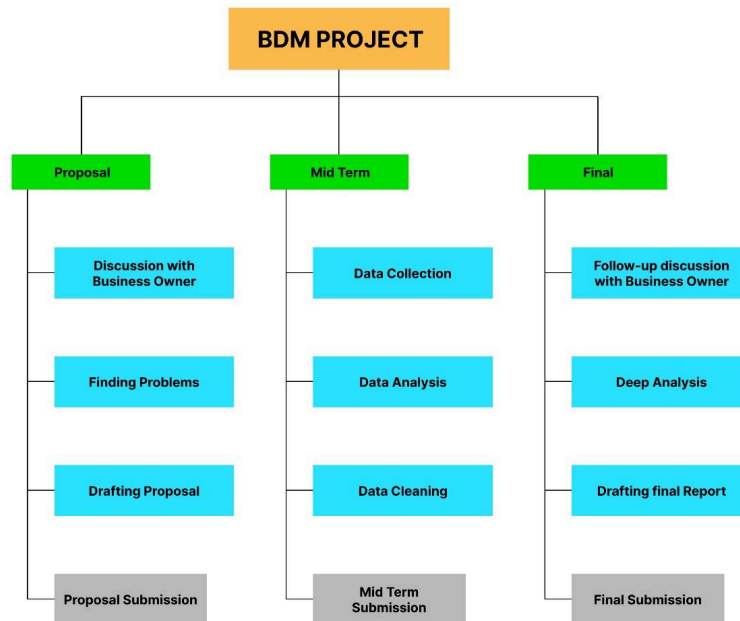


Figure 6.1: Work Breakdown Structure using flowchart

6.2 **Gantt chart:** In the below Figure 2 is the gantt chart along with the required excel WBS

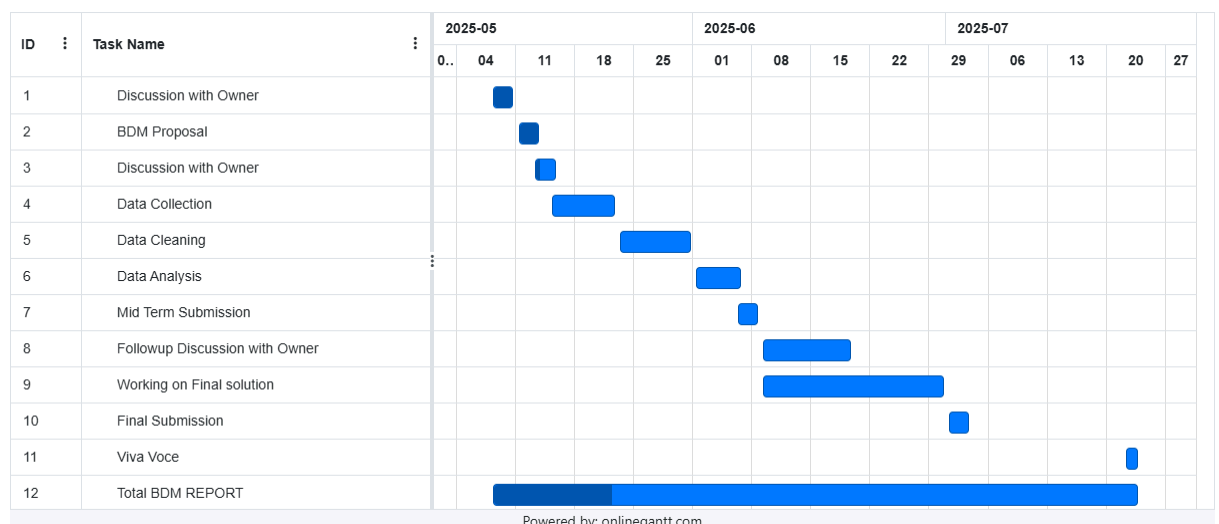


Figure 6.2.1: Gantt Chart using the website onlinegantt.com

	A	B	C	D
1	Task	Start	End	Duration
2	Discussion with Owner	2025-05-08	2025-05-10	3
3	BDM Proposal	2025-05-11	2025-05-13	3
4	Discussion with Owner	2025-05-13	2025-05-15	3
5	Data Collection	2025-05-15	2025-05-22	8
6	Data Cleaning	2025-05-23	2025-05-31	9
7	Data Analysis	2025-06-01	2025-06-06	6
8	Mid Term Submission	2025-06-06	2025-06-08	3
9	Followup Discussion with Owner	2025-06-09	2025-06-19	11
10	Working on Final solution	2025-06-09	2025-06-30	22
11	Final Submission	2025-07-01	2025-07-03	3
12	Viva Voce	2025-07-22	2025-07-23	2
13	Total BDM REPORT	2025-05-08	2025-07-23	77

Figure 6.2.2: WBS using excel used to create Gantt Chart

7 Expected Outcome

The goal of this project is to help Diabetes Care Medical and General Stores use data smarter to improve daily operations and make future growth possible.

1. By analyzing profit and sales data, we can figure out how much money the store can safely reinvest. With 20–25 days of supplier credit and good monthly profits, expanding to a second branch becomes realistic if finances are better planned.
2. Sales trends and ABC analysis will help the store stock the right products in the right quantities. This avoids overstocking slow-moving items and running out of high-demand ones.
3. By calculating safety stock and reorder points, stock management becomes less of a guess and more data-backed, leading to fewer wasted products and more satisfied customers.
4. These changes will also reduce the manual workload for staff, making their everyday tasks simpler and more accurate.
5. Combining better financial planning and smarter stock decisions will make the store run smoother and give it a stronger base to grow in the future.

In conclusion, this project is expected to improve efficiency, reduce errors, and support sustainable growth for the store.